Disarming anti-phage defence systems with synthetic inhibitory protein binders

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Introduction

- Bacterial anti-phage defence systems limit the therapeutic potential of bacteriophages (phages)
- Inspired by anti-CRISPR proteins, this study explores synthetic inhibitors to block key components of two defence systems: Shango and VP1840
- BindCraft, a tool that leverages AI and deep learning, predicts protein-protein interactions and designs inhibitory binders using models from AlphaFold2 and ProteinMPNN

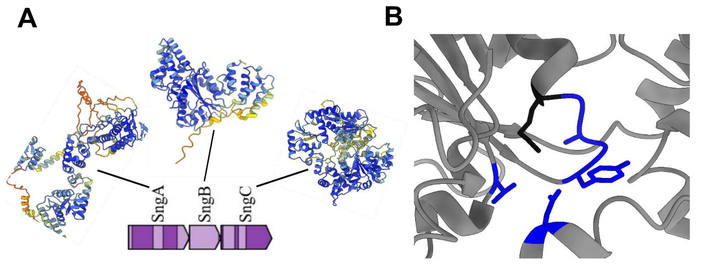


Figure 1. Shango defence system. **A**: Shango comprised of 3 gene system, SngA (TerB-like domain), SngB (ATPase), and SngC (Helicase). **B**: Active site of SngB, Lys65 (Black), various amino acids (Blue)

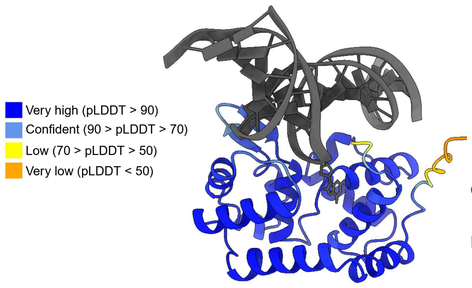


Figure 2. VP1840 defence system. VP1840 binds to DNA (dark grey) and degrades it

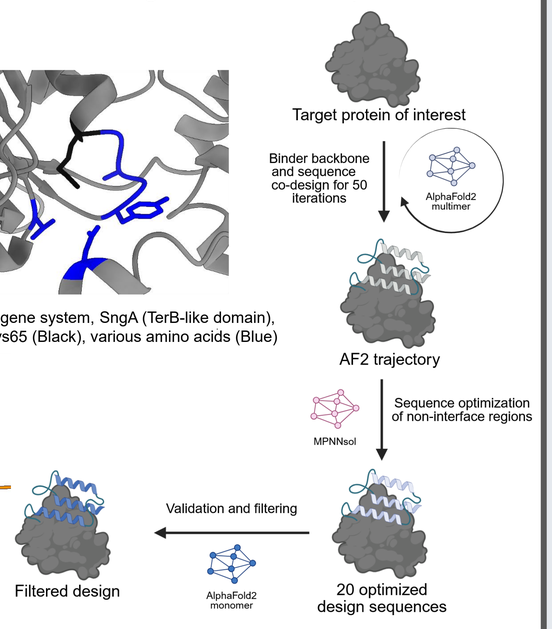


Figure 3. BindCraft pipeline. Schematic representation of the BindCraft design pipeline.

Objective

Design a proof-of-concept AI pipeline for generating protein binders that inhibit anti-phage defences, with the goal of engineering phages to carry these binders, enabling them to bypass bacterial immunity and improve infectivity.

Experimental Design

Dry Lab:

Target Identification

Structural Modelling via AlphaFold3

Binder Design via BindCraft

Chain: "A"

Target hotspot residues:
- K65 (SngB)
- N/A (Vp1840)

Lengths: [65, 150]

Number of final designs: 100

BindCraft

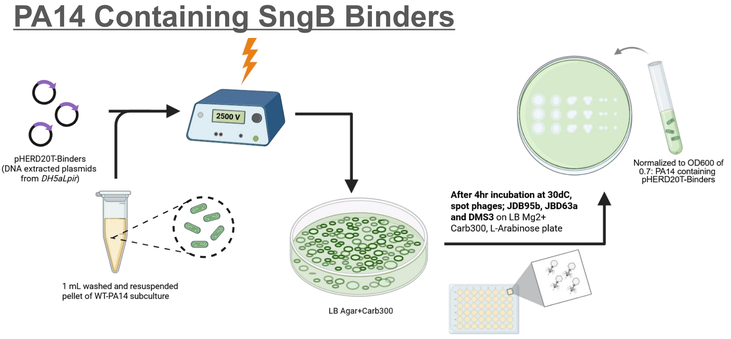
Filtered Design

Post-processing and Filtering

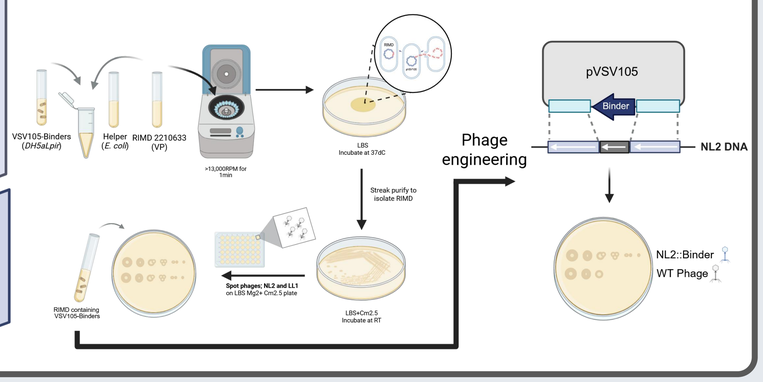
Top 20/100 binders (ipTM-ranked), filtered for interface overlap at predicted active sites

Wet Lab:

PA14 Containing SngB Binders



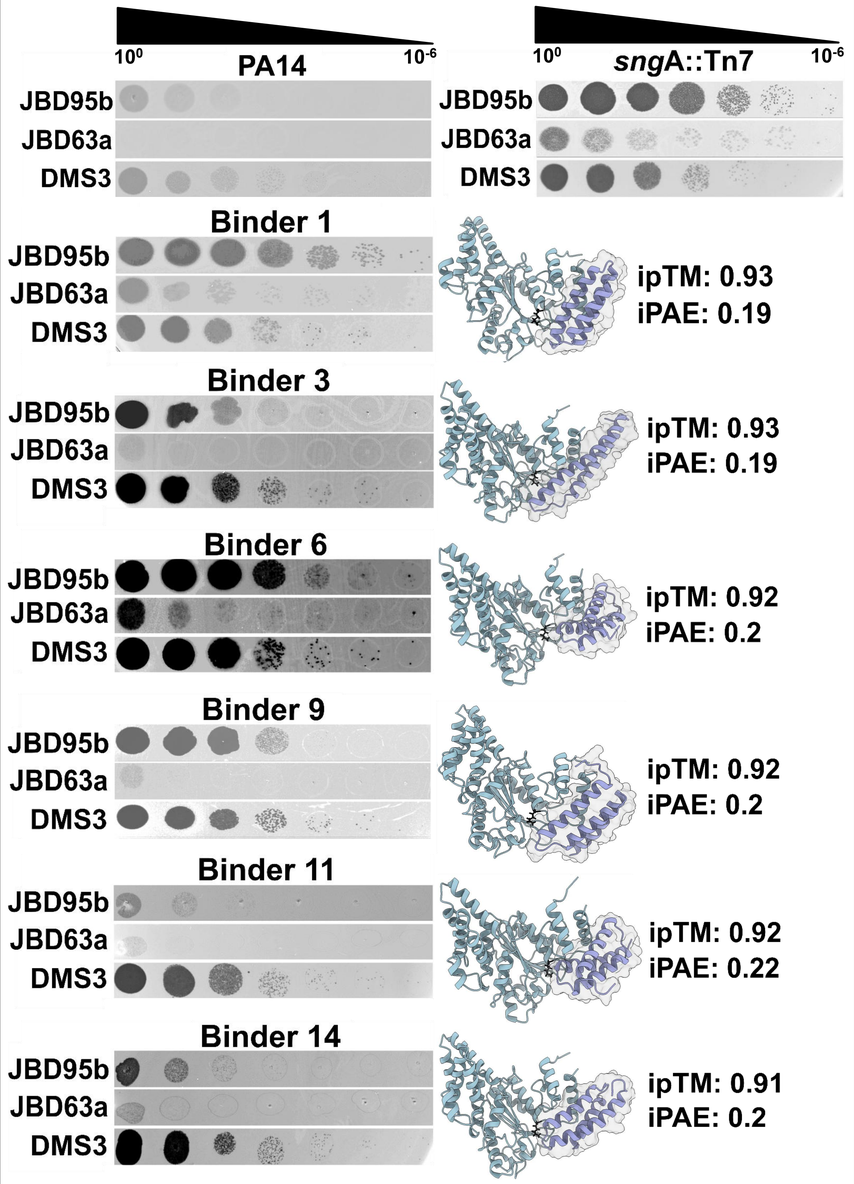
RIMD 2210633 Containing VP1840 Binders



Results

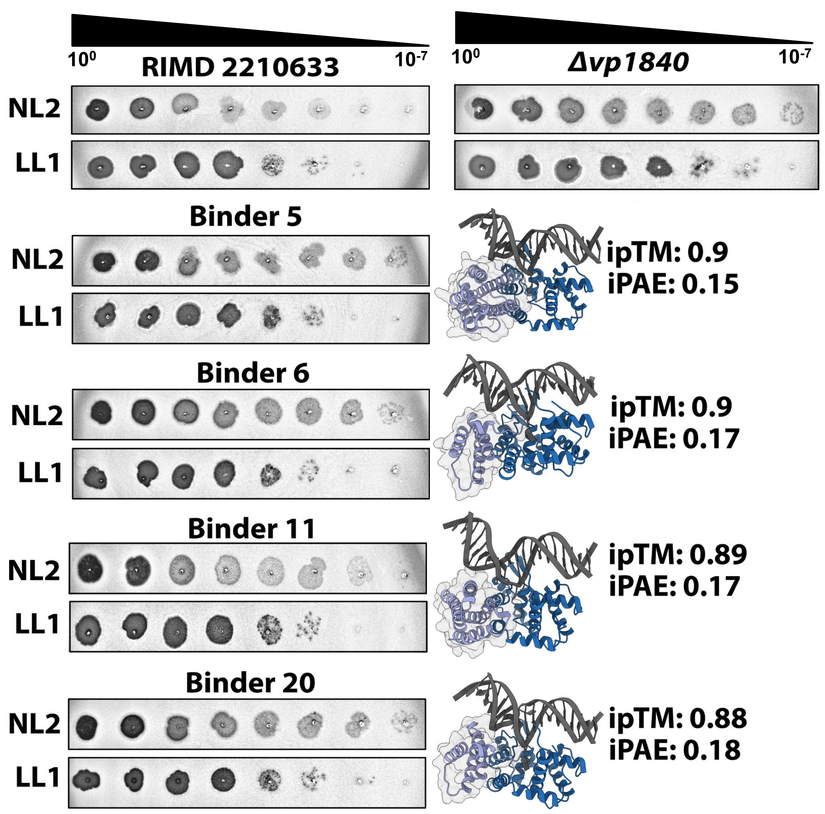
SngB Binders

Binder | SngB | Lys65



VP1840 Binders

Binder | VP1840 | DNA



Conclusions

- AI-designed binders can effectively inhibit bacterial defense systems with target and context-specific activity
- Six out of 10 SngB binders restored JBD95b phage infectivity in *P. aeruginosa* PA14
- SngB Binders 9, 11, and 14 reduced plaque formation by ~100-fold in JBD63a
- Four out of five VP1840 binders successfully restored plaque formation

Future Directions

- Engineer NL2 phage mutants carrying VP1840 binders through Gibson assembly and allelic exchange
- Design NT-CRISPR gRNA targets, assemble into *V. parahaemolyticus RIMD 2210633* to selectively screen mutant phages
- Perform molecular dynamics simulations to analyze protein–binder interaction mechanisms
- Validate binder-defence system interactions through co-purification protein assays and test binding affinities via fluorescence polarization assays
- Refine binders by iterating designs using ProteinMPNN to enhance sequence performance

References

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6. Images created with BioRender.com